Final Exam Solutions:

1. **(Probabilistic Baristas; 25 pts)**

   A coffee shop serves one random milk alternative each day: Walnut (with probability 0.5), Pistachio (w.p. 0.4), and Soy (w.p 0.1).

   Suppose that you have an minor allergy to Walnuts; upon consumption you have a 30% chance of developing an allergic reaction. The other types of milk do not affect you in any way.

   In this problem we assume you visit the coffee shop and purchase a cup of coffee with milk every day.

   (a) What is the probability that you have an allergic reaction on a given day?

   (b) Given that you do not have an allergic reaction on a given day, what is the probabilities that the coffee shop served walnut milk on that day?

   (c) What is the expected number of times during one week (7 days) that you consume walnut milk, but do not get an allergic reaction?

   Hint: Start by building a Bernoulli random variable for the event that you consume walnut milk ***and*** *have no reaction on any one day.*

(a) By law of total probability:

$$\mathbb{P}(\text{Allergy}) = \mathbb{P}(\text{Allergy} \mid \text{Walnut}) \cdot \mathbb{P}(\text{Walnut}) + \mathbb{P}(\text{All.} \mid \text{Pistachio}) \cdot \mathbb{P}(\text{Pistachio}) + \mathbb{P}(\text{All.} \mid \text{Soy}) \cdot \mathbb{P}(\text{Soy})$$

Plug in from problem

$$= 30\% \cdot 50\% + 0\% \cdot 40\% + 0\% \cdot 10\%$$

$$\boxed{\mathbb{P}(\text{Allergy}) = 15\%}$$

(b) Use Bayes' Rule:

$$\mathbb{P}(\text{Walnut} \mid \text{No Allergy}) = \frac{\mathbb{P}(\text{No Allergy} \mid \text{Walnut}) \cdot \mathbb{P}(\text{Walnut})}{\mathbb{P}(\text{No Allergy})}$$

$$= \frac{(1 - 30\%) \cdot 50\%}{1 - 15\%}$$

$$\boxed{\mathbb{P}(\text{Walnut} \mid \text{No Allergy}) = \frac{7}{17} \approx 0.41}$$

(c) On day $i = 1, 2, \dots 7$, let

$$X_i = \begin{cases} 1 & \text{if Walnut \& No Allergy} \\ 0 & \text{o/w} \end{cases}$$

$$\mathbb{P}(X_i = 1) = \mathbb{P}(\text{Walnut} \cap \text{No Allergy})$$

$$= \mathbb{P}(\text{No Allergy} \mid \text{Walnut}) \cdot \mathbb{P}(\text{Walnut})$$

$$= (1 - 30\%) \cdot 50\% = \underline{35\%}$$

$$\mathbb{E}\left[\begin{matrix}\text{\# of times in 7 days} \\ \text{with Walnut \& No Allergy}\end{matrix}\right] = \mathbb{E}\left[\sum_{i=1}^{7} X_i\right]$$

Linearity of $\mathbb{E}[\cdot]$

$$= \sum_{i=1}^{7} \mathbb{E}[X_i]$$

X_i's are identically distributed

$$= 7 \cdot \mathbb{E}[X_1]$$

$$\boxed{\mathbb{E}\left[\sum_{i=1}^{7} X_i\right] = 7 \cdot 35\% = \underline{2.45}}$$

2. **(Joint Random Variables and Risk; 25 pts)**
In a simplified version of the board game 'Risk', two players each roll a fair 6-sided die to determine the outcome of a battle.

Let A denote the value of the attacking player's die roll, and D the defending player's die roll. Define a new random variable W as follows:

$$W = \begin{cases} 1 & \text{if } A > D \\ 0 & \text{otherwise.} \end{cases}$$

Namely, W equals one if and only if the attacking player wins – when their die roll A is *strictly* greater than the defender's die roll D.

In this problem, assume that both players' die rolls are independent, and have equally likely outcomes.

(a) Compute the joint probability mass function of A and D.

(b) Compute the probability mass function of W.

(c) Compute the joint probability mass function of D and W. Be careful to note that D and W will *not* be independent, and their ranges are distinct.

(d) Compute the conditional probability mass function of W given $D = d$, for each possible $d \in \{1, 2, 3, 4, 5, 6\}$.
Your answer should contain six different conditional pmf's for W, corresponding to each possible value of d.

(e) Let $W_1, W_2, W_3, \ldots$ denote the random outcome of several different such battles. Assume that all die rolls are independent, so that $W_1, W_2, \ldots,$ are independent as well.
Let N be the random variable indicating the first time the attacking player wins; i.e., if we observe $W_1 = 0, W_2 = 0, W_3 = 1$, then $N = 3$ since the first win occurred on the third battle.
Compute both: i) the distribution of N, and ii) $\mathbb{E}[N]$, the expected number of battles until the attacking player wins.

(a) Since A, D each are uniform rv's with range $\{1, 2, \ldots 6\}$, then

$$p_A(a) = 1/6, \quad a = 1, 2, \ldots 6$$

$$p_D(d) = 1/6, \quad d = 1, 2, \ldots 6$$

By independence of the die rolls, we have

$$p(a, d) = p_A(a)\, p_D(d)$$

Hence: $\boxed{p(a, d) = \mathbb{P}(A = a, D = d) = 1/36, \text{ for any } a, d \in \{1, 2, \ldots 6\}}$

(b) 36 total outcomes for the pair (A, D)

- 6 are equal rolls $\to W = 0$
- 15 have $D > A \to W = 0$
- 15 have $A > D \to W = 1$

So:

$$\boxed{\begin{aligned} \mathbb{P}(W=1) &= \frac{15}{36} = \frac{5}{12} \approx 42\% \\ \mathbb{P}(W=0) &= \frac{6+15}{36} = \frac{7}{12} \approx 58\% \end{aligned}}$$

(c) $\mathbb{P}(W=1, D=d) = \mathbb{P}(A>d, D=d)$

$= \mathbb{P}(A>d)\cdot\mathbb{P}(D=d)$) By A, D independent

$= \underline{\left(1-\frac{d}{6}\right)\cdot\frac{1}{6}}$

$\mathbb{P}(W=0, D=d) = \mathbb{P}(A\le d, D=d)$

$= \mathbb{P}(A\le d)\cdot\mathbb{P}(D=d)$

$= \underline{\frac{d}{6}\cdot\frac{1}{6}}$

Together:

$$\boxed{\mathbb{P}(W=w, D=d) = \begin{cases} \frac{1}{6}-\frac{d}{36} & \text{if } w=1 \\ \frac{d}{36} \end{cases}, \text{ for any } d=1,2,\ldots 6}$$

(d) $\mathbb{P}(W=w \mid D=d) = \dfrac{\mathbb{P}(W=w, D=d)}{\mathbb{P}(D=d)}$ $\}=1/6$, for any $d=1\ldots 6$

$$\therefore \underline{\mathbb{P}(W=w \mid D=d)} = \begin{cases} 1-d/6 & \text{if } w=1 \\ d/6 & \text{if } w=0 \end{cases}, \text{ for } d=1\ldots 6$$

(e) We know from (b) that $W\sim \text{Bernoulli}(5/12)$.

By construction, $\boxed{N\sim\text{Geometric}(5/12)}$. Hence, $\boxed{E[N]=\frac{1}{5/12}=\frac{12}{5}=2.4}$

"# of Bernoulli trials until 1st success"

3. **(MLE of Lognormal Distribution; 25 pts)**

A random variable X has a *lognormal distribution* with parameters μ and $\sigma^2 > 0$ (denoted $X \sim \text{LogNorm}(\mu, \sigma^2)$) if its density function equals:

$$f(x; \mu, \sigma^2) := \frac{1}{x\sqrt{2\pi\sigma^2}} \cdot \exp\left(-\frac{(\log(x) - \mu)^2}{2\sigma^2}\right) \qquad \forall x > 0,$$

where $\log(x)$ is the natural logarithm (base e), and the density equals zero for any $x \leq 0$.

In this question, we will build and analyze maximum likelihood estimators for the lognormal family of distributions.

Note: Despite what the notation may suggest, μ is **not the mean** of X, and σ^2 is **not the variance** of X.

(a) First, we will prove that if $X \sim \text{LogNorm}(\mu, \sigma^2)$, then $\log(X) \sim \text{Norm}(\mu, \sigma^2)$.

Show that for any $t \in \mathbb{R}$, we have:

$$\mathbb{P}(\log(X) \leq t) = \int_{-\infty}^{t} \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left(-\frac{(v-\mu)^2}{2\sigma^2}\right) dv.$$

Hint: Express the left-hand side as an integral over the density above, and perform a change of variables in the integral with $v = \log(x)$.

(b) Let $X_1, ..., X_n \overset{iid}{\sim} \text{LogNorm}(\mu, \sigma^2)$.

Compute the likelihood function $\mathcal{L}(\mu, \sigma^2; X_1, ..., X_n)$, and the log-likelihood function $L(\mu, \sigma^2; X_1, ..., X_n)$.

(c) Find $\hat{\mu}_{MLE}(X_1, ..., X_n)$, the maximum likelihood estimator for μ.

Hint: Be sure to differentiate the (log)-likelihood function with respect to μ *in order to get the MLE for* μ *itself. Your answer shouldn't depend on* σ^2.

(d) For any value of n, what is the **exact** distribution of $\hat{\mu}_{MLE}(X_1, ..., X_n)$? Is it an unbiased estimator for μ? Justify your answer (you should not need a proof for this).

(e) Find $\hat{\sigma}^2_{MLE}(X_1, ..., X_n)$, the maximum likelihood estimator for σ^2, by solving

$$\frac{\partial L}{\partial(\sigma^2)}\left(\hat{\mu}_{MLE}, \hat{\sigma}^2_{MLE}\right) = 0.$$

Be sure to take the partial derivative with respect to (σ^2), and not σ.

Is it an unbiased estimator for σ^2?

(a) $P(\log(X) \leq t) = \mathbb{P}(X \leq e^t)$

$$= \int_0^{e^t} \frac{1}{x} \cdot \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(\log(x)-\mu)^2} dx$$

Integrate pdf from 0 to e^t

Substitution: $v = \log(x)$, $dv = \frac{1}{x} dx$

Upper: $x = e^t \iff v = t$ Lower: $x = 0 \iff v = -\infty$

$$\boxed{P(\log(X) \leq t) = \int_{-\infty}^{t} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(v-\mu)^2\right) \cdot dv \quad \text{as desired.}}$$

(b) $\mathcal{L}(\mu, \sigma^2; X_1 \ldots X_n) = \prod_{i=1}^n f(X_i; \mu, \sigma^2)$ by $X_1 \ldots X_n$ iid $\text{LogNormal}(\mu, \sigma^2)$

$$= \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \frac{1}{X_i} \cdot \exp\left(-\frac{1}{2\sigma^2}\left(\log(X_i) - \mu\right)^2\right)$$

Simplifying

$$\boxed{\mathcal{L}(\mu, \sigma^2; X_1 \ldots X_n) = (2\pi\sigma^2)^{-\frac{n}{2}} \cdot \left(\prod_{i=1}^n \frac{1}{X_i}\right) \exp\left[-\frac{1}{2\sigma^2} \sum_{i=1}^n \left(\log(X_i) - \mu\right)^2\right]}$$

$$L(\mu, \sigma^2; X_1 \ldots X_n) = \log\left(\mathcal{L}(\mu, \sigma^2; X_1 \ldots X_n)\right)$$

$$\boxed{L(\mu, \sigma^2; X_1 \ldots X_n) = -\frac{n}{2}\log(2\pi\sigma^2) - \left[\sum_{i=1}^n \log(X_i)\right] - \frac{1}{2\sigma^2}\left[\sum_{i=1}^n \left(\log(X_i) - \mu\right)^2\right]}$$

(c) Solve: $0 = \frac{\partial L}{\partial \mu}(\hat{\mu}_{MLE})$

$$0 = 0 - 0 - \frac{1}{2\sigma^2} \sum_{i=1}^n 2 \cdot \left(\log(X_i) - \hat{\mu}_{MLE}\right) \cdot (-1)$$

$$\iff 0 = \sum_{i=1}^n \left(\log(X_i) - \hat{\mu}_{MLE}\right)$$

$$\iff \boxed{\underline{\hat{\mu}_{MLE} = \frac{1}{n} \sum_{i=1}^n \log(X_i)}}$$

(d) From (a), we know that $\log(X_i) \sim N(\mu, \sigma^2)$

So, $\hat{\mu}_{MLE}$ is the average of n iid $N(\mu, \sigma^2)$ rv's!

$\Rightarrow \boxed{\hat{\mu}_{MLE} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \text{ \underline{exactly}}}$ (since $\log(X_1), \ldots \log(X_n) \overset{iid}{\sim} N(\mu, \sigma^2)$)

$\boxed{\text{Hence, } \mathbb{E}[\hat{\mu}_{MLE}] = \mu, \text{ \underline{Unbiased}}}$

(e) Solve: $0 = \frac{\partial L}{\partial(\sigma^2)}(\hat{\mu}_{MLE}, \hat{\sigma}^2_{MLE})$

$$0 = \frac{\partial}{\partial(\sigma^2)}\left[-\frac{n}{2}\log(2\pi\hat{\sigma}^2_{MLE}) - \left[\sum_{i=1}^{n}\log(X_i)\right] - \frac{1}{2\hat{\sigma}^2_{MLE}}\left[\sum_{i=1}^{n}\left(\log(X_i)-\hat{\mu}_{MLE}\right)^2\right]\right]$$

$$= -\frac{n}{2}\cdot\frac{1}{\hat{\sigma}^2_{MLE}} - 0 + \frac{1}{2(\hat{\sigma}^2_{MLE})^2}\left[\sum_{i=1}^{n}\left(\log(X_i)-\hat{\mu}_{MLE}\right)^2\right]$$

$$\Leftrightarrow n\cdot\hat{\sigma}^2_{MLE} = \sum_{i=1}^{n}\left(\log(X_i)-\hat{\mu}_{MLE}\right)^2$$

$$\boxed{\therefore \hat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{i=1}^{n}\left(\log(X_i)-\hat{\mu}_{MLE}\right)^2}$$

This is a biased estimator. $\sigma^2 = Var(\log(X_1))$, and when dividing by n (without knowing $\mu = E[\log(X_1)]$) we have a biased estimator for variance.

In contrast, $\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(\log(X_i)-\hat{\mu}_{MLE}\right)^2$

or

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}\left(\log(X_i) - \underbrace{\mu}_{= E[\log(X_1)]}\right)^2$$

would be unbiased estimators for σ^2.

4. **(Unconfident Intervals; 25 pts)**
Recall the Exponential distribution with parameter $\lambda > 0$ has the following pdf:

$$f(x;\lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0, \\ 0 & x < 0. \end{cases}$$

Suppose we have $2n$ random variables $X_1, ..., X_n, Y_1, ..., Y_n$ with i.i.d. Exponential(λ) distributions, where the value of λ is unknown. However, due to some horrific clerical error, we can only observe $Z_1, ..., Z_n$, where

$$Z_i = \lambda(X_i - Y_i + 1),$$

for any $i = 1...n$ and λ is the same unknown parameter above.

In this question we will explore some properties of the random variables $Z_1, ..., Z_n$ for making inferences about λ.

(a) What are $\mathbb{E}[Z_1]$ and $\mathrm{Var}(Z_1)$?

(b) Consider the following estimator for λ:

$$S_{40} = \frac{1}{40}\sum_{i=1}^{40} Z_i.$$

What are $\mathbb{E}[S_{40}]$ and $\mathrm{Var}(S_{40})$? Justify your answer.
Hint: Each Z_i is independent, and $\mathbb{E}[S_{40}]$ should depend on λ.

(c) Find a two-sided 95% confidence interval for $\mathbb{E}[S_{40}]$. Be sure to justify each step.

(d) Find an expression for C, depending on S_{40}, for which $\mathbb{P}(\lambda \geq C) \approx 0.05$.

(e) Consider the confidence interval found in part (c). Would the width of the interval shrink, expand, or remain the same when the true λ parameter increases? Why?

(a) $\mathbb{E}[Z_1] = \mathbb{E}[\lambda(X_1 - Y_1 + 1)]$) Linearity

$$= \lambda \underbrace{\mathbb{E}[X_1]}_{=1/\lambda} - \lambda \underbrace{\mathbb{E}[Y_1]}_{=1/\lambda} + \lambda$$

$$\boxed{\mathbb{E}[Z_1] = \lambda}$$

$\mathrm{Var}(Z_1) = \mathrm{Var}(\lambda X_1 - \lambda Y_1 + \lambda)$) By X_1, Y_1 indep.

$$= \lambda^2 \underbrace{\mathrm{Var}(X_1)}_{=1/\lambda^2} + \lambda^2 \underbrace{\mathrm{Var}(Y_1)}_{=1/\lambda^2}$$

$$\boxed{\mathrm{Var}(Z_1) = 2}$$

(b) $Z_1, \ldots Z_n$ are iid with mean λ and variance 2.

So: $$\boxed{\begin{aligned} \mathbb{E}[S_{40}] &= \mathbb{E}[Z_1] = \lambda \\ Var(S_{40}) &= \tfrac{1}{40} \cdot Var(Z_1) = \tfrac{1}{20} \end{aligned}}$$

(c) By CLT, $\dfrac{\sqrt{40}(S_{40} - \lambda)}{\sqrt{2}} \sim N(0,1)$

So: $$\mathbb{P}\left(\underbrace{-1.96}_{-z_{2.5\%}} \le \frac{\sqrt{40}(S_{40}-\lambda)}{\sqrt{2}} \le \underbrace{1.96}_{z_{2.5\%}} \right) = 95\%$$

Re-arranging to isolate λ

$\iff$ $\boxed{\underline{S_{40} \pm 1.96 \cdot \frac{\sqrt{2}}{\sqrt{40}}}}$ is a 95% CI for $\mathbb{E}[S_{40}] = \lambda$!

(d) Need: $\mathbb{P}(\lambda \ge C) = 5\%$

$\iff (-\infty, C]$ is a 95% CI for λ

So: $$\boxed{C = S_{40} + \underbrace{1.65}_{z_{5\%}} \cdot \frac{\sqrt{2}}{\sqrt{40}}}$$ $\left[\text{Comes from } \mathbb{P}\left(\frac{\sqrt{40}(S_{40}-\lambda)}{\sqrt{2}} \ge -1.65 \right) = 95\% \right]$

(e) The <u>width remains identical</u> at $2 \cdot 1.96 \cdot \frac{\sqrt{2}}{\sqrt{40}}$, regardless of the value of λ. This is because $Var(Z_1)$ doesn't depend on λ.

5. **(Hypothesis Tests on the Median; 25 pts)**

Let $f(\cdot)$ be a probability density function describing some population of interest, and $F(\cdot)$ the corresponding CDF. Define the unknown population parameter $\theta := F^{-1}\left(\frac{1}{2}\right)$ as the *median* of this distribution. I.e., θ is the 50th percentile of a random variable with density $f(\cdot)$; an example is visualized here:

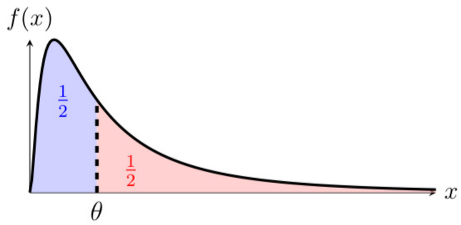

Figure 1: An example density $f(\cdot)$ and its median θ. Both the red and blue shaded areas equal $\frac{1}{2}$.

Let $X_1, ..., X_n$ be i.i.d. random variables, each with marginal pdf $f(\cdot)$, and let $H_0 : \theta = \theta_0$, $H_A : \theta > \theta_0$ be two hypotheses. Define new random variables $Y_1, ..., Y_n$ such that

$$Y_i = 1\{X_i \geq \theta_0\}, \qquad \forall i = 1...n.$$

In other words, $Y_i = 1$ if and only if $X_i \geq \theta_0$; otherwise $Y_i = 0$.

(a) What is the joint distribution of $Y_1, ..., Y_n$ under H_0?

(b) Assume that $n > 40$.
What is the distribution of $\bar{Y}_n := \frac{1}{n}\sum_{i=1}^{n} Y_i$ under H_0?

(c) In the setting of part (b), use $Y_1, ..., Y_n$ to build a test statistic and corresponding rejeciton region for a test of size α with H_0 and H_A as specified above. Be sure to justify why your hypothesis test has the desired size of α.

(d) Assume that both $n > 40$, and that $H_A : \theta > \theta_0$ is true. As a result, now that θ_0 is not the true median, $\mathbb{P}(X_i \geq \theta_0) = 1 - F(\theta_0) = \frac{1}{2} + \delta$ for some $\delta > 0$.
What is the new joint distribution of $Y_1, ..., Y_n$ and $\bar{Y}_n$ now that H_A is true?
Hint: Be very careful when applying the CLT here.

(e) In the setting of part (d), what is the new distribution of the test statistic in part (c)? Use this to compute the probability of a Type II Error. You answer should depend on δ, n, and can be left in terms of any common CDF's.

(a) Under H_0, $\mathbb{P}(Y_i=1) = \mathbb{P}(Y_i=0) = 1/2$

So $Y_i \sim \text{Bernoulli}(1/2)$

Since $X_1 ... X_n$ are iid, then $Y_1 ... Y_n$ are iid.

$\boxed{\therefore Y_1 ... Y_n \overset{iid}{\sim} \text{Bernoulli}(1/2) \text{ under } H_0}$

(b) By CLT, since $\mathbb{E}[Y_1] = 1/2$ and $Var(Y_1) = 1/4$, (and $n > 40$)

then $\dfrac{\sqrt{n}(\bar{Y}_n - 1/2)}{1/2} \sim N(0,1)$ approximately.

Thus: $\boxed{\bar{Y}_n \sim N\left(\frac{1}{2}, \frac{1}{4n}\right) \text{ under } H_0}$

(c) $\underline{T(Y_1 \ldots Y_n)} = \frac{\sqrt{n}(\bar{Y}_n - 1/2)}{1/2}$ $(\sim N(0,1)$ under $H_0)$

$\underbrace{\underline{R = [z_\alpha, \infty)}}$ "T too large" $\iff$ "Over $1/2$ of X_i's above θ_0"

$\iff$ "True median should be above θ_0"

$\mathbb{P}(T \in R \mid H_0)$

$= \mathbb{P}(N(0,1) \geq z_\alpha) = \underline{\alpha}$

(d) Now, $\mathbb{P}(Y_i = 1) = \mathbb{P}(X_i \geq \theta_0) = \frac{1}{2} + \delta$, since H_0 is false

So: $\boxed{Y_1 \ldots Y_n \overset{iid}{\sim} \text{Bernoulli}\left(\frac{1}{2} + \delta\right)}$

Note: $\text{Var}(Y_1) = \left(\frac{1}{2} + \delta\right)\left(1 - \left(\frac{1}{2} + \delta\right)\right) = \frac{1}{4} - \delta^2$

So by CLT: $\frac{\sqrt{n}\left(\bar{Y}_n - \left(\frac{1}{2} + \delta\right)\right)}{\sqrt{\frac{1}{4} - \delta^2}} \sim N(0,1)$ Notice both $E[\bar{Y}_n]$ and $\text{Var}(\bar{Y}_n)$ changed!

$\boxed{\therefore\ \bar{Y}_n \sim N\left(\frac{1}{2} + \delta, \frac{1}{n}\left(\frac{1}{4} - \delta^2\right)\right)}$

(e) From part (d), $\bar{Y}_n \sim N\left(\frac{1}{2} + \delta, \frac{1}{n}\left(\frac{1}{4} - \delta^2\right)\right)$

So: $T(Y_1 \ldots Y_n) = \frac{\sqrt{n}\left(\bar{Y}_n - \frac{1}{2}\right)}{1/2} = \frac{\sqrt{\frac{1}{4} - \delta^2}}{1/2} \cdot \frac{\sqrt{n}\left(\bar{Y}_n - \frac{1}{2} - \delta + \delta\right)}{\sqrt{\frac{1}{4} - \delta^2}}$

$$= \frac{\sqrt{\frac{1}{4} - \delta^2}}{1/2} \cdot \left[\underbrace{\frac{\sqrt{n}\left(\bar{Y}_n - \frac{1}{2} - \delta\right)}{\sqrt{\frac{1}{4} - \delta^2}}}_{\sim N(0,1) \text{ under } H_A} + \frac{\sqrt{n}\,\delta}{\sqrt{\frac{1}{4} - \delta^2}}\right]$$

$\boxed{\Rightarrow T(\ldots) \sim N\left(2\delta\sqrt{n}, 1 - 4\delta^2\right) \text{ under } H_A}$

$$\mathbb{P}(\text{Type II Error}) = \mathbb{P}\left(N(2\delta\sqrt{n},\ 1-4\delta^2) \le z_\alpha\right)$$ "Don't reject, but should"

$$= \mathbb{P}\left(N(0,\ 1-4\delta^2) \le z_\alpha - 2\delta\sqrt{n}\right)$$

$$= \mathbb{P}\left(N(0,1) \le \frac{z_\alpha - 2\delta\sqrt{n}}{\sqrt{1-4\delta^2}}\right)$$

$$\boxed{\mathbb{P}(\text{Type II Error}) = \Phi\left(\frac{z_\alpha - 2\delta\sqrt{n}}{\sqrt{1-4\delta^2}}\right)}$$

6. **(Regressing on a Binary Variable; 25 pts)**
Consider a linear regression setting, with a random sample of i.i.d. data points $(X_i, Y_i)_{i=1}^n$ from the model $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$ for unknown β_0 and β_1. Assume that for all i, the error $\epsilon_i \sim \mathrm{N}(0, \sigma_\epsilon^2)$ is independent of X_i.

The independent variable X_i is *binary*, namely it can take on only values of 0 or 1.

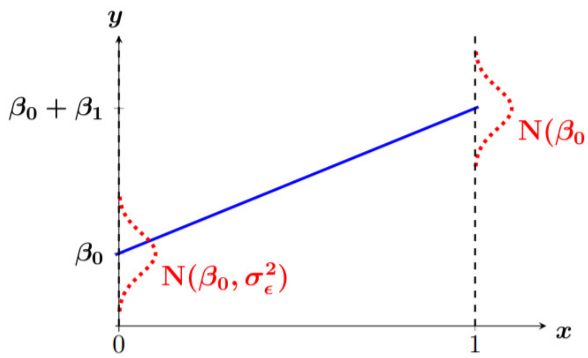

Figure 2: Linear Regression with a binary independent variable $X \in \{0, 1\}$. The blue line is the true population regression line $y = \beta_0 + \beta_1 x$. The red dotted bell curves show the conditional density of Y given either $X = 0$ or $X = 1$.

You may use without proof the fact that we can express both:

$$S_{XY} = \left(\sum_{i=1}^n X_i Y_i\right) - n\bar{X}\bar{Y}, \qquad S_{XX} = \left(\sum_{i=1}^n X_i^2\right) - n\bar{X}^2,$$

where $\bar{X} := \frac{1}{n}\sum_{i=1}^n X_i$ and $\bar{Y} := \frac{1}{n}\sum_{i=1}^n Y_i$ are the random sample means.

(a) Let $\bar{Y}_{(0)}$ be the mean of all Y_i's for which $X_i = 0$.

Show that
$$\bar{Y}_{(0)} = \frac{1}{n - n\bar{X}} \sum_{i=1}^n (1 - X_i) Y_i.$$

(b) Let $\bar{Y}_{(1)}$ be the mean of all Y_i's for which $X_i = 1$.

Show that
$$\bar{Y}_{(1)} = \frac{1}{n\bar{X}} \left(\sum_{i=1}^n X_i Y_i\right).$$

(c) Show that the linear regression estimator for the slope equals $\hat{\beta}_1 = \bar{Y}_{(1)} - \bar{Y}_{(0)}$.

(d) Show that the linear regression estimator for the intercept equals $\hat{\beta}_0 = \bar{Y}_{(0)}$.

(a) Since $X_i = 0$ or $X_i = 1$, $\sum_{i=1}^n X_i = n\bar{X} = \#$ of i's with $X_i = 1$

$\Rightarrow n - n\bar{X} = \#$ of i's with $X_i = 0$

And: $(1 - X_i) Y_i = \begin{cases} Y_i & \text{if } X_i = 0 \\ 0 & \text{o/w} \end{cases}$

So: $\boxed{\bar{Y}_{(0)} = \dfrac{\text{Sum of } Y_i\text{'s with } X_i = 0}{\#\text{ of } i\text{'s with } X_i = 0} = \dfrac{1}{n - n\bar{X}} \sum_{i=1}^n (1 - X_i) Y_i}$ as desired

(b) Use same idea as (a). $X_i \cdot Y_i = \begin{cases} Y_i & \text{if } X_i = 1 \\ 0 & \text{o/w} \end{cases}$

$$\bar{Y}_{(1)} = \frac{\text{Sum of } Y_i\text{'s if } X_i = 1}{\#\text{ of } Y_i\text{'s with } X_i = 1} = \frac{1}{\underbrace{\sum_{i=1}^{n} X_i}_{= n\bar{X}}} \sum_{i=1}^{n} X_i Y_i$$

(c)
$$\bar{Y}_{(1)} - \bar{Y}_{(0)} = \frac{1}{n\bar{X}}\left(\sum_{i=1}^{n} X_i Y_i\right) - \frac{1}{n - n\bar{X}}\left(\sum_{i=1}^{n} (1-X_i) Y_i\right)$$

$$= \frac{1-\bar{X}}{n\bar{X} - n\bar{X}^2}\left(\sum_{i=1}^{n} X_i Y_i\right) - \frac{\bar{X}}{n\bar{X} - n\bar{X}^2}\left(\sum_{i=1}^{n} (1-X_i) Y_i\right)$$

$$= \frac{1}{n\bar{X} - n\bar{X}^2}\left[(1-\bar{X}) \sum_{i=1}^{n} X_i Y_i - (\bar{X}) \sum_{i=1}^{n} (1-X_i) Y_i\right]$$

$$= \sum_{i=1}^{n} X_i Y_i \cancel{- \bar{X} \sum_{i=1}^{n} X_i Y_i} - \bar{X} \cdot \underbrace{\sum_{i=1}^{n} Y_i}_{= n \cdot \bar{Y}} \cancel{+ \bar{X} \sum_{i=1}^{n} X_i Y_i}$$

$$= \sum_{i=1}^{n} X_i Y_i - n \cdot \bar{X}\bar{Y}$$

$$= S_{XY}$$

$$= \frac{S_{XY}}{n\bar{X} - n\bar{X}^2}$$

$$] = \sum_{i=1}^{n} X_i - n\bar{X}^2$$

$$= \sum_{i=1}^{n} X_i^2 - n\bar{X}^2 \quad \} \text{ Since } 0^2 = 0,\ 1^2 = 1$$

$$= S_{XX}$$

So: $\bar{Y}_{(0)} - \bar{Y}_{(1)} = \dfrac{S_{XY}}{S_{XX}} = \hat{\beta}_1$ as desired. ☒

(d) We know: $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$, by def'n of $\hat{\beta}_0$

$$= \frac{1}{n} \sum_{i=1}^{n} Y_i - \left[\bar{Y}_{(1)} - \bar{Y}_{(0)}\right] \cdot \bar{X}$$

$$= \sum_{i=1}^{n} (1-X_i) Y_i + \sum_{i=1}^{n} X_i Y_i$$

$$= (n - n\bar{X}) \cdot \bar{Y}_{(0)} + (n\bar{X}) \cdot \bar{Y}_{(1)} \quad \} \text{ from (a), (b)}$$

$$= (1 \cancel{- \bar{X}}) \cdot \bar{Y}_{(0)} \cancel{+ \bar{X} \cdot \bar{Y}_{(1)}} \cancel{- \bar{X} \cdot \bar{Y}_{(1)}} \cancel{+ \bar{X} \cdot \bar{Y}_{(0)}}$$

So: $\hat{\beta}_0 = 1 \cdot \bar{Y}_{(0)}$ as desired. ☒